Fiscal Year:	FY 2025	**Task Last Updated:**	FY 12/03/2024
PI Name:	Hammer, Bruce Ph.D.		
Project Title:	Study of Mammalian Pluripotent Stem Cells in Microgravity		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology		
Space Biology Cross-Element Discipline:	(1) Developmental Biology		
Space Biology Special Category:	(1) Cell Culture		
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Zip Code:	55455-3007	**Congressional District:**	5
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2014 Space Biology Flight NNH14ZTT001N
Start Date:	11/01/2014	**End Date:**	09/30/2023
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
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Flight Program:	ISS		
Flight Assignment:	ISS NOTE: End date changed to 09/30/2023 per NSSC (Ed., 8/25/22) NOTE: Extended to 1/31/2022 per F. Hernandez/ARC (Ed., 7/27/21) NOTE: Extended to 6/30/2021 per F. Hernandez/ARC (Ed., 1/18/21) NOTE: Extended to 10/31/2020 per F. Hernandez/ARC and NSSC information (Ed., 6/18/20) NOTE: Extended to 6/30/2020 per NSSC information (Ed., 1/29/2020) NOTE: Extended to 11/1/2019 per F. Hernandez/ARC (Ed., 11/6/18) NOTE: Extended to 11/1/2018 per F. Hernandez/ARC (Ed., 10/21/16)		
Key Personnel Changes/Previous PI:	The Principal Investigator reports that Louis Kidder, Ph.D. is no longer with the project (Ed., 11/17/22).		
COI Name (Institution):	Dutton, James Ph.D. (University of Minnesota)		
Grant/Contract No.:	NNX15AB38G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	We are investigating the effect of gravity on fundamental properties of mammalian stem cells during differentiation of 3-D cultures of induced pluripotent stem (iPS) cells. Experiments aboard the ISS, STS and ground-based simulations have demonstrated that microgravity influences gene expression, cell proliferation and differentiation in stem cells. However, the mechanism behind these observations is not clearly understood. Oct4 is a transcription factor necessary for maintaining pluripotency in mammalian stem cells. It is expressed ubiquitously in the early embryo, pluripotent ES and iPS cells and is rapidly down regulated during differentiation. In this study we will investigate how exposure to microgravity fundamentally alters the regulation of Oct4, and how these changes can affect the timing, progression and outcomes of cell differentiation. Our laboratory has created an Oct4:CreER mTmG transgenic mouse that, for the first time, allows lineage tracing of Oct4 expression in stem cells and their progeny. We will use iPS cells derived from this model to determine the influence of microgravity on the loss of pluripotency and differentiation. We are using magnetic levitation, a unique ground-based simulation of orbital free fall, to augment and compare the findings from the Spaceflight Experiment Phase (micro-15). Unfortunately astronaut handling of the samples aboard the ISS resulted in only a few EB's returned rather than the expected hundreds of EB's which resulted in a loss of science. We propose to investigate the effect of gravity on the timing and spatial arrangement of the loss of Oct4 expression in differentiating iPS cell aggregates via magnetic levitation. Using iPS cells derived from the Oct4:CreER mTmG mice and timed tamoxifen additions, it is possible over time to distinguish cells still expressing Oct4 (i.e., expressing GFP), from those that have lost Oct4 expression (expressing RFP). We will also examine the effect of gravity on gene expression in cohorts of Oct4 expressing and non-expressing cells during differentiation by comparing the results of ground-based experiments to those conducted on orbit. Finally, we will explore mechanisms behind the effect of microgravity on both Oct4 gene regulation and control of downstream gene expression by Oct4. This work will determine the effect of spaceflight on changes in Oct4 gene expression during differentiation of pluripotent stem cells and the consequences of these changes on differentiation outcomes. This will increase our understanding of fundamental stem cell behavior in microgravity.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Gravity likely influences organismal development but little is currently understood about its specific influence on stem cell differentiation. We propose to utilize a novel system to investigate how microgravity fundamentally changes the timing and spatial arrangement of the loss of Oct4 gene expression during the differentiation of stem cells. We expect this to have consequences on differentiation outcomes. Our results from this study will ultimately have a direct impact on improving the translation of human stem cell based treatments. Cell manufacturing in microgravity may speed-up the rate of iPSC (induced Pluripotency Stem Cell) differentiation, thereby reducing the time and cost to obtain a therapeutic dose of cells. If this can be done in orbit and replicated on Earth with magnetic levitation, this will have significant commercialization possibilities.
Task Progress:	Final report for the NASA Task Book, per the Principal Investigator (Ed., 12/3/24): Science was severely impacted by the low number of stem cell embryoid bodies (EBs) recovered in each bag / time point. • This was primarily due to having a crew member who was not trained in handling biological samples carry out the study. The cells were improperly thawed and withdrawal and replacement of media was done too aggressively. This occurred because the astronaut who was originally assigned to the experiment had to attend to a more urgent on station issue. • Another unexpected complicating factor was the presence of a large volume of air / foam throughout the culture media which appears to be a consequence of the microgravity environment. Lessons Learned • Increase time allowed for crew training and activities or delay experiment execution when other mitigating on-station circumstances occur. • Have the science team test extended times for centrifugation and post centrifugation activities in order to document impact on cells. • More clearly understand the impact of bubbles during and after centrifugation. • If safety allows it, clip the bags immediately following the end of centrifugation prior to transport to the Life Sciences Glovebox (LSG) • Use anti-foaming agents in future studies.
Bibliography Type:	Description: (Last Updated:)